



Reddit Hyperlink Network: Conflict, Connectivity, and Community Structure

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How does conflict between subreddits shape Reddit's broader network structure?



Sub-questions:



Do negative links increase separation between groups?



Are hostile links more common between clusters?



Which subreddits act as bridges?



Do some subreddits attract more negative attention?



Does this change over time

Dataset Overview

Dataset: Reddit Hyperlink Network (SNAP)

Time Range: Jan 2014 – Apr 2017

Nodes: 35,776 subreddits

Edges: 137,821 directed hyperlinks

Edge Sign/ Sentiment Distribution:

- +1 = positive/neutral = 265,491 = 92.65%
 - -1 = negative = 21,070 = 7.35%
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Who Links and Who Gets Linked

Top Source Subreddits (most outgoing links):

- subredditdrama
- circlebroke
- shitliberalssay
- outoftheloop
- copypasta

Top Target Subreddits (most incoming links):

- askreddit
 - iama
 - pics
 - writingprompts
 - videos
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Overall Network Structure

- **Strongly Connected Components:**
 - Total: 24,071
 - Largest: 11,564 nodes
- **Weakly Connected Components:**
 - Total: 497
 - Largest: 34,671 node

**Global
Metrics
(Largest
WCC)**

Nodes: 34,671

Edges: 123,570

Avg shortest path: 3.87

Diameter: 10

Global clustering coefficient: 0.1862

Density: 0.0002

ER vs BA vs Reddit Network

- **Erdős–Rényi (ER) Random Graph**

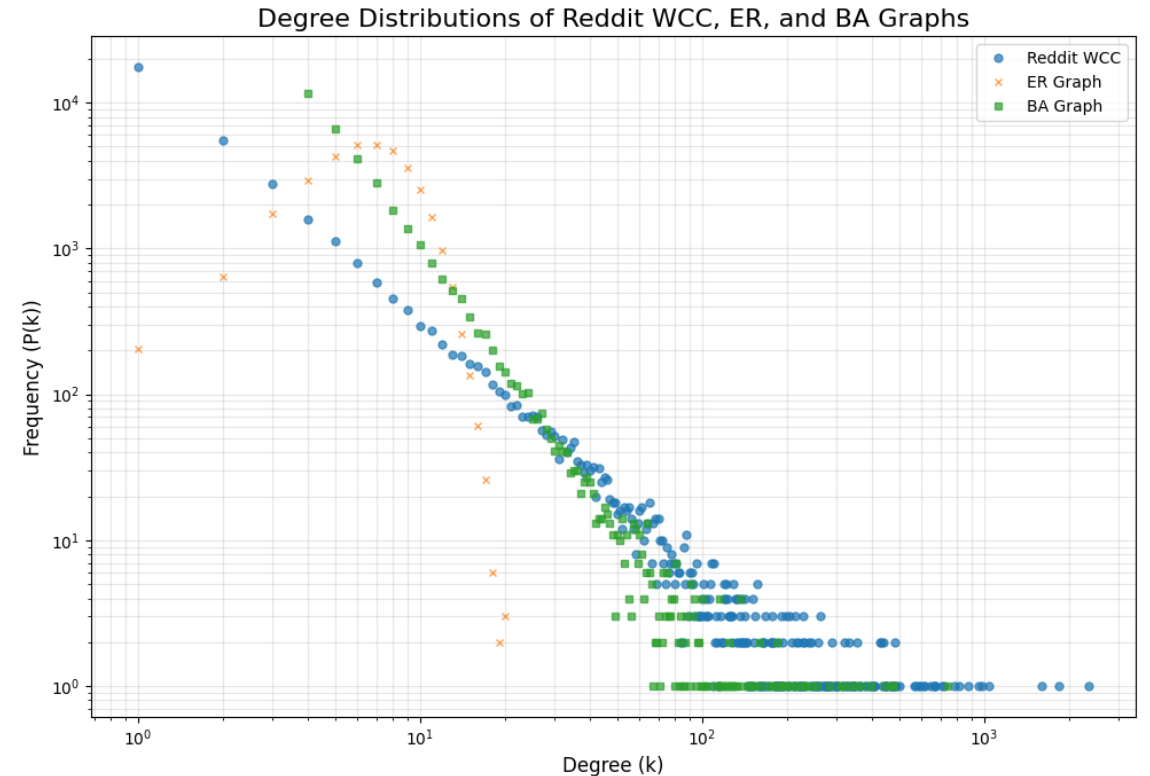
- Very low clustering (0.0002)
- Longer paths (avg = 5.56)
- Structure is mostly random

- **Barabási–Albert (BA) Scale-Free Graph**

- Slightly higher clustering (0.0023)
- Shorter paths (avg = 4.30)
- Has hubs but still low clustering

- **Real Reddit Network**

- Much higher clustering (0.1862)
- Short paths (avg ~3.87)
- Strong community structure
- However, between ER and BA graphs, the Reddit degree distribution resembles the BA graph the most



Centrality Findings

- **In-Degree (who gets linked to):** askreddit, iama, pics
- **Out-Degree (who links out):** subredditdrama, copypasta, drama
- **Betweenness:** askreddit, iama, subredditdrama
- **Closeness:** askreddit, iama, subredditdrama

Early vs Late Network

Early Period Vs Late Period

16,781 nodes → 28,469 nodes

53,280 edges → 98,308 edges

Clustering: 0.1564 → 0.1648

Avg path: 4.06 → 3.99

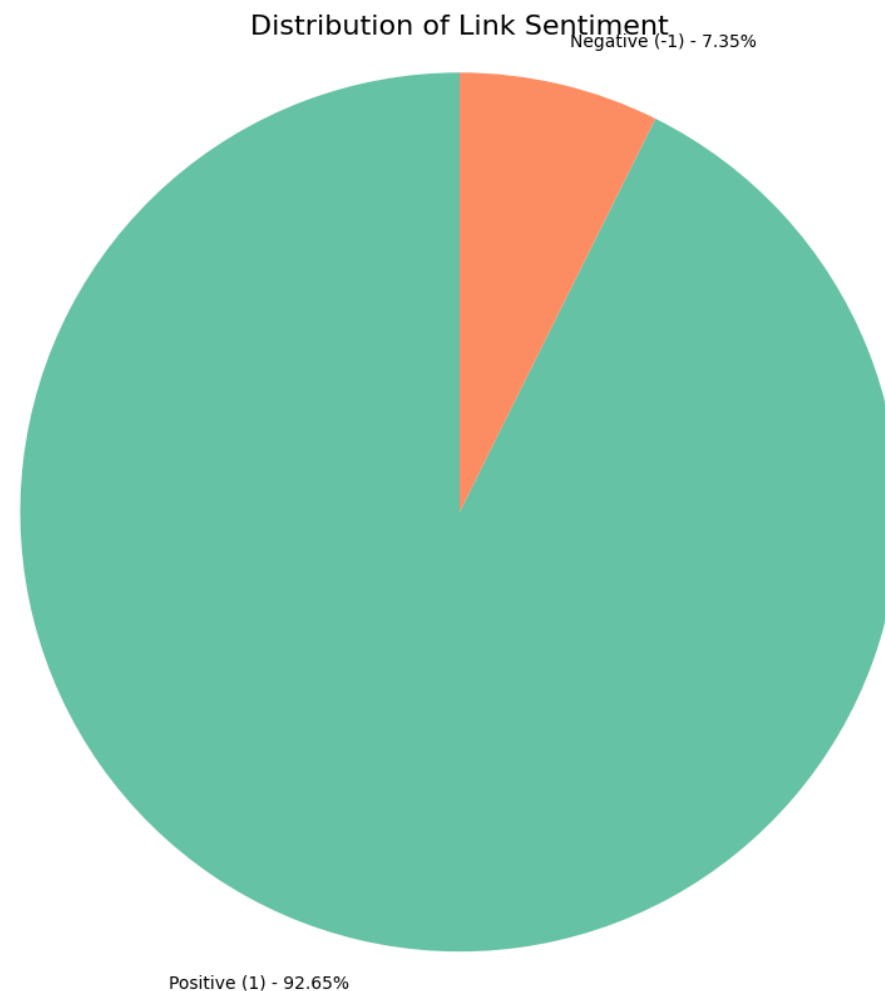
Positive vs Negative Networks

- **Positive Network:**

- 35,010 nodes
- 130,513 edges
- Clustering: 0.1774

- **Negative Network:**

- 6,370 nodes
- 14,513 edges
- Clustering: 0.0945



Negative Network Communities



371 communities



Largest community: 638 nodes



Modularity: 0.5130



High-betweenness subs spread across many communities

Assortativity

- Degree assortativity: -0.0930
- Community assortativity: 0.5526

Results

- Negative links do not drive clustering
- Negative links often bridge communities
- Positive links form the dense core of Reddit
- Conflict highlights boundaries but also connects distant groups

THANK YOU

